Site Audit Module – Project Guide

BestServe Pest Management · ps-ops · Status as of 27 Oct 2026 · For answering questions about the project

In one line: A supervisor visits a customer site, inspects it area by area on the mobile app (photos + voice notes in Kannada/Hindi/English, translated to English by Gemini), signs it off, submits it, and the system automatically makes a professional PDF report that can be downloaded or shared from the phone or the web.

1. What problem it solves

- Site audits were done on paper / WhatsApp photos and typed up later. That was slow, inconsistent and hard to track.
- Now the supervisor records everything on the phone at the site, in their own language, and the customer gets a standard branded PDF report the same day.
- Every finding has a risk level, a deadline ("fix within") and actions for the customer and for BestServe, so follow-up is clear.
- Admins and supervisors can see all audits on the web, filter them and download reports.

2. Who uses it (roles and permissions)

Role	What they can do	Permissions
Supervisor	Start audits, inspect areas, add photos and voice, sign off, submit, view and download reports (mobile and web)	VIEW_AUDIT, CREATE_AUDIT, UPDATE_AUDIT, SUBMIT_AUDIT
Admin	Everything a supervisor can do, on all audits. Also holds the reserved permissions for later features	All 8: the four above + AMEND_AUDIT, GENERATE_AUDIT_REPORT, OVERRIDE_AUDIT_RISK, MANAGE_AUDIT_SETTINGS
Other roles	No access (decided: only supervisor and admin)	None

A supervisor only sees audits inside their own scope (their branch / company / site, using the existing `user_scopes`). If they open an audit outside their scope they get "not found", so the system never reveals that it exists.

3. The full flow, step by step

#	Step	What happens	Where
1	Start audit	Supervisor taps **+ Start New Audit**, searches the customer/site, picks the date (Today / Tomorrow / Pick date), audit types (multi-select, default General Pest Control), target pests (Rodents, Cockroaches, Mosquitoes, Flies, or "+ Other" typed), and optionally the site contact person (name, role, phone). The supervisor is the auditor automatically. An audit number like **BSP-AUD-2026-**	Mobile / Web

		0005 is created. If the site had a previous audit, its areas can be copied in one tap.	
2	Audit Areas list	Shows every area with its number, risk, status and first photo. The next unfinished area is highlighted with **Start / Continue**. Progress shows like "4 / 8". Buttons: **+ Add Area** and **Review →**.	Mobile
3	Inspect an area – Details	Area name and zone (optional). Then one or more **findings**: "What did you find?" (optional – if blank, the risk category is used as the title), risk category (chips, or "+ Other" typed), risk level (Low / Medium / High / Critical), **Fix within** (Now / 7 / 15 / 30 days / Other – always chosen by the supervisor), and the due date is shown.	Mobile
4	Evidence (photos)	Big **Take Photo** button or choose from gallery. Photos upload straight away; the first one is the ★ **Cover** photo for the report. Long-press to remove; tap a failed photo to retry.	Mobile
5	Observation (voice or type)	Pick the language (ಕನ್ನಡ Kannada / हिंदी Hindi / English / Auto), tap the mic and speak, or "type instead". The recording is uploaded and Gemini returns professional English in a few seconds. The supervisor can edit the English, play the original recording, or retake it.	Mobile
6	Actions	Two optional fields: **Customer action** (e.g. "Hospital should…") and **BestServe action**. Each can be voice or typed. If not recorded by voice, "Quick add" chips insert ready sentences (Clear debris, Seal gaps, Bait stations…).	Mobile
7	Save area	**Save Draft** keeps work without completing. **Save Area** marks it completed and goes straight back to the area list ("✓ area saved", next area highlighted). If something required is missing, it stays a draft and the app shows exactly what to fill.	Mobile / Web
8	Review & Submit	A checklist: green ✓ for what is done, orange ! with **Fix →** for what is missing. **Report received by** (pre-filled from the site contact) and the **Next audit** (2 weeks / 1 month / 3 months / pick). Optional sign-off details (designation, remarks, which report pages). **Submit Audit** locks the audit.	Mobile / Web
9	PDF report	Made automatically in the background a few seconds after submit. The "Audit submitted" screen shows **Pest Risk Audit Report · N pages · ready** with **Download PDF** and **Open / Share (WhatsApp, Email)**. On the web: **Download Report (PDF)**.	Mobile / Web

Statuses

Audit status	Meaning
IN_PROGRESS	Being filled in; can be edited (shown as a draft on the phone).

SUBMITTED (shown as "Completed")	Locked; can be viewed, not changed; report is made.
AMENDING	Reserved for re-opening a submitted audit (the screen to start an amendment is not built yet).
CANCELLED	Reserved.

Area status	**Meaning**
NOT_STARTED	Added (e.g. copied from the last audit) but nothing entered. Can be left out at submit.
IN_PROGRESS	Saved as a draft.
COMPLETED	Saved with everything required.

4. Risk levels and deadlines

Level	**Colour**	**Report timeline bucket**	**Typical fix within**
Critical	Dark red	Immediately	1 day / Now
High	Orange	Within 7 days	7 days
Medium	Yellow	Within 30 days	30 days
Low	Light green	Ongoing	Monitor

The supervisor always chooses "Fix within" (it does not change by itself when the risk level changes). Due date = audit date + days. The audit's **overall risk** is the highest risk of its findings.

5. Voice notes and translation (Gemini)

- The phone records audio (m4a) and uploads it. The audio is always kept as evidence, even if translation fails.
- The **server** (never the phone) sends it to Google **Gemini**, so the API key stays secret on the server.
- Gemini writes the exact transcript in the spoken language, then a **professional English** translation with the **same meaning** – nothing added, removed or analysed; only filler words (e.g. "sir", "that's it") are dropped.
- Models: main `gemini-3.8-flash`, backup `gemini-3.6-flash`. The backup starts only if the main one has not answered in 6 seconds. Each call times out after 20 seconds; a busy/network failure is retried after 2 seconds; a background job every 30 seconds picks up anything left.
- If the recording has no clear speech, it says so and offers **Retake**.

- Normal translation time: about **4–14 seconds**.

Important limit: The current Gemini API key is on Google's **free tier: 20 requests per day per model** (resets around 12:30 PM India time). That is roughly 10–20 recordings a day in total. When it is used up, the app says "Today's translation limit is used up. Type the English text, or try again tomorrow." **To fix: enable billing** on the key's Google project in Google AI Studio (paid Tier 1 starts instantly). No code change and no new key needed. Cost is small per recording. The exact paid daily limit is shown in AI Studio → Rate limits.

6. Photos

- Stored in **MinIO** (our file storage, bucket `ps-ops`) under `audits/…`; database keeps only the reference.
- The app shows small thumbnails (made by the server) so lists load fast; the full photo is used in the PDF.
- Removing a photo hides it from the report; the original is kept as evidence (soft delete).
- Every upload has an id made on the phone, so a retry after a network drop never creates a duplicate.

7. The PDF report

16:9 landscape, BestServe branding (logo and fonts already used for quotations). Made with **pdfkit** on the server. Stored in MinIO at `audits/{audit}/reports/v1.pdf`; the data used is also frozen as a JSON snapshot, so the report never changes later.

Page	Content	When shown
Cover	Customer, site address, date, report no., auditor, site contact, audit type, pests, overall risk badge, cover photo	Always
Summary	Counts by risk level, areas inspected, findings register (no., area, finding, risk)	Always (continues on more pages if long)
Risk chain	Shelter → Burrows → Entry with the findings in each	Option (on by default)
Risk by zone	Zones with counts per risk level	Option
This week's priority	High and Critical findings with due dates and actions	Option, and only if any High/Critical
Finding pages	One per finding: 2 photos, zone/area/category/fix-within/due, then Observation, Customer action, BestServe action. Extra photo page if 3+ photos	Always

Action timeline	Immediately / Within 7 days / Within 30 days / Ongoing	Always
IPM cycle	Inspect → Identify → Prevent → Control → Monitor → Review	Option
Sign-off	Audited by, site contact, received by, next audit, remarks, BestServe contact	Always

Every page footer: "Bestserve Pest Management · {customer}", page n of N, report version. A typical small audit is about 9–10 pages; it is made in a few seconds. If making it fails it retries up to 3 times, then shows "Try again".

Sharing on mobile: "Download PDF" saves to **Downloads/BestServe** (WhatsApp/email can attach from there). "Open / Share" opens the phone's PDF viewer, which has its own Share button. A one-tap "Share on WhatsApp" button would need a new library (`react-native-share`) – not installed, needs approval.

Made simple on purpose

- Most choices are taps on chips, not typing: date, audit type, pests, risk category, fix within, zones used before, quick-add action sentences.
- Removed to save time: the separate "Area Saved" screen, the "Risk type" field, and site notes on the phone. Site contact, finding title and the two actions are optional.
- "Report received by" is filled in from the site contact automatically.

Battery and data

- The app checks for the translation every 3 seconds, then every 6 seconds, and stops completely when the app is in the background. Same for the "report being made" check, which also stops once the report is ready.
- No ticking timers while waiting; photos are taken at 1600 px (enough for the report), so uploads are smaller and faster.

8. Rules the system enforces

- **Required to start:** customer/site, at least one audit type, at least one pest (typed if "Other"). Site contact is optional.
- **Required for a finding:** risk category (typed if "Other"), risk level, fix within. The title is optional.
- **To complete an area:** area name and the observation. Customer and BestServe actions are optional.
- **To submit:** at least one area; all started areas completed (never-started areas can be left out with a switch); all voice notes accepted/typed; "report received by" entered.
- **Locking:** a submitted audit cannot be changed.
- **Two people editing:** every save carries a version number; if someone else saved first, the app says so instead of overwriting (no lost work). If a different user continues an audit, it is logged as a hand-over in History.
- **History:** every important action (start, save, submit, hand-over…) is recorded with who and when.

9. Screens

Mobile app (supervisor → Audits tab, green theme)

Screen	Purpose
Audit Home	Greeting, "N audits in progress · N due today", Start New Audit, Continue where you left off (Continue → Area NN), Recent submitted audits
Start New Audit	Customer/site search, date chips, type and pest chips, contact, notes
Audit Areas	All areas, next-area highlight, progress, Add Area / Review, report card when submitted
Area Inspection (4 steps)	Details → Evidence → Observation → Actions; Save Area returns to the area list
Sign-off	Received by, designation, next audit date, remarks, optional report pages
Review & Submit	Checklist with Fix →, received by, next audit, submit
Audit Submitted	Big ✓, report card with Download / Open-Share, Back to Audits

Web (Admin and Supervisor menus → Site Audits)

Page	Purpose
Audits list	Search, From/To date, status and auditor filters; table of report no., customer/site, date, auditor, areas, status; Continue / View
Audit detail	Tabs: Summary, Areas & Findings, Photos & Evidence, Actions, Sign Off, History; Edit, Review & Submit, Download Report (PDF)

10. Technology and where the code is

Part	Technology	Main files
Backend API	Node 20, Express 5, MySQL 8.4, MinIO, Redis, node-cron, in Docker	`backend/src/routes/audits.routes.js` (all /api/audits endpoints), `utils/audits.js`, `utils/auditTree.js`

Voice translation	Google Gemini REST API	`services/AuditVoice.service.js`, `jobs/auditQueue.cron.js`
PDF report	pdfkit + sharp (photo resize)	`services/AuditReport.service.js`, `utils/auditReportPdf.js`
Mobile app	React Native 0.87 (Android), react-navigation, nitro-sound (audio), image-picker, blob-util (PDF)	`mobile/src/audits/` (screens, api.ts, media.tsx, report.tsx, ui.tsx)
Web	React 19 + Vite	`frontend/src/pages/audits/`
Docs	Word	`docs/Site Audit - Architecture.docx`, `API Payloads.docx`, `Table Design.docx`, `Mobile Design.png`, this guide

Database (10 new tables)

Table	Holds
audit_risk_levels	Critical/High/Medium/Low with default days, colour, timeline bucket
audits	One row per audit: number, customer, site, date, types, pests, auditor, contact, status, overall risk, report options
audit_areas	Areas of an audit: number, name, zone, status
audit_findings	Findings: title, category, risk, fix within, due date, observation, actions
audit_finding_photos	Photo references (MinIO key, size, order, cover, include in report)
audit_voice_notes	Recordings: language, transcript, English draft, accepted text, status, attempts
audit_recommendations	Action-timeline lines (table ready; editing screen not built – timeline is built from findings)
audit_signoffs	Received by, designation, next audit date, remarks

audit_reports	PDF versions: status, pages, size, checksum, snapshot
audit_history	Audit trail of every action

Main API endpoints (/api/audits)

lookups · site-options · area-suggestions · auditors · summary · list · create · get · edit header · add areas · save area · delete area · sign-off · readiness · submit · reports (list / file / retry) · files (upload / view) · photos (edit / delete) · voice (get / retry / accept) · history. Full payloads: docs/Site Audit - API Payloads.docx.

11. What is done and what is pending

Phase	Scope	Status
1. Foundation	Tables, start/save/sign-off/submit, mobile and web screens	Done
2. Photos & voice	Uploads, Gemini translation, retake, playback	Done
3. PDF report	Option B PDF, auto after submit, download/share on mobile and web	Done
Design refresh	Mobile screens redesigned to the approved board (chips, checklist, big buttons)	Done
4. Offline & handover	Work with no network, sync later	Not started – needs a new library (approval)
Later	Amend a submitted audit; edit action-timeline lines; risk override; notifications ("report ready"); one-tap WhatsApp share; admin settings	Not built (permissions already reserved)

12. Known issues and risks (be ready for these)

- **Gemini free tier** – 20 requests/day per model. Needs billing turned on for real use.
- **Fallback model quality** – the backup model is sometimes less accurate; the supervisor can always edit the English or retake.
- **Security housekeeping** – the repo root contains a Maps API key text file and an Android signing key (`bestserve-key.jks`); an SMTP password was visible in a screenshot. Recommended: rotate the SMTP password and Gemini key, and keep secrets out of the repository.
- **Emulator only tested so far** – needs testing on real phones, on weak networks, and with large audits (30+ findings).

- **Existing web bug (not from this module)** – in the Supervisor web menu, "Dashboard" always looks selected.
- **schema.prisma is outdated** – the backend uses raw SQL; the new tables are not in Prisma.

13. Questions your manager may ask – with answers

Q. How long does one audit take to record?

About the same time as walking the site. Each area is: type a title, tap chips, take photos, speak the observation and actions. Translation takes a few seconds. No typing up afterwards.

Q. Does the supervisor need to know English?

No. They can speak Kannada or Hindi; Gemini turns it into professional English with the same meaning. They can check and edit it before saving.

Q. What if the internet is weak at the site?

Uploads retry safely (no duplicates), and drafts are saved on the server. Full offline mode (Phase 4) is not built yet.

Q. What if translation fails or is wrong?

The original audio is always kept. The supervisor can retry, retake, or type the English. Nothing is lost.

Q. When does the customer get the report?

Immediately after submit – the PDF is ready in a few seconds and can be downloaded or shared from the phone, or downloaded from the web.

Q. Can a submitted audit be changed?

No, it is locked so the report always matches the data. An "amend" flow is planned (status and permission exist) but not built yet.

Q. Who can see an audit?

Admins (all) and supervisors (only within their own branch/company/site scope). Other roles have no access.

Q. Where are photos and recordings stored? Are they safe?

In our own MinIO storage, not on Google. Files are only served through the backend after a login and permission check. Only the audio is sent to Gemini for translation.

Q. What does it cost to run?

Everything runs on our existing servers. The only external cost is Gemini – free now (20/day limit), a small pay-per-use amount once billing is enabled.

Q. Can two supervisors work on the same audit?

Yes, one after the other (hand-over is logged). If both save at the same moment, the second one is told to reload instead of overwriting.

Q. What happens if the PDF fails to generate?

Only the report fails – the audit data is safe. It retries automatically up to 3 times, then shows "Try again".

Q. What is left to do?

Turn on Gemini billing, test on real phones, then decide on: offline mode, amending submitted audits, editing the action timeline, report-ready notifications and one-tap WhatsApp share.

14. How to run it (developer notes)

- **Backend:** `cd infra && docker compose up -d` (MySQL, Redis, MinIO, backend on port 3000). Backend settings come from `backend/.env` (Gemini key, MinIO bucket `ps-ops`).
- **Web:** `cd frontend && npm run dev` → log in as admin or supervisor → Site Audits.
- **Mobile (Android emulator):** start Metro (`cd mobile && npx react-native start`), start the Pixel_8 emulator, `adb reverse tcp:8081 tcp:8081`, open the BestServe app, log in as a supervisor → Audits tab. For voice on the emulator: `adb emu avd hostmicon`. If the emulator freezes, reboot it.